Probability & Statistics

Books on probability and statistics
https://www.quora.com/What-are-some-good-books-for-learning-probability-and-statistics

Introduction to Probability Theory

Probability Definitions
Probability of an event is the measure of the likelihood that the event will occur (in some experiment).

Probability is a number between 0 and 1.
But sometimes people express probability using percents.

0 – event is impossible
0.5 – the event is just as likely to occur as it is to not occur
1 – event will certainly happen, 100%.

The sum of probabilities of all outcomes of an experiment should always be 1.

Outcomes can be combinations of individual events.

For example, if we throw one dice cube – there are 6 possible outcomes (1,2,3,4,5,6), each one has same probability – 1/6.

If we throw 2 cubes, there are 6*6=36 possible combinations.
Probability of each combination is the same – 1/36.

If we define an outcome as a sum of numbers on top side of cubes after the throw, then there are 11 possible outcomes:

Sum	Combinations	Number of combinations	Probability
2	1+1	1	1/36
3	1+2,2+1	2	2/36
4	1+3,2+2,3+1	3	3/36
5	1+4,2+3,3+2,4+1	4	4/36
6	1+5,2+4,3+3,4+2,5+1	5	5/36
7	1+6,2+5,3+4,4+3,5+2,6+1	6	6/36
8	2+6,3+5,4+4,5+3,6+2	5	5/36
9	3+6,4+5,5+4,6+3	4	4/36
10	4+6,5+5,6+4	3	3/36
11	5+6,6+5	2	2/36
12	6+6	1	1/36
Total		36	1

From the table above you see how some outcomes can result from more combinations, thus have higher probability of happening.

Note: Events can be mutually exclusive, or not.
Also, events may be independent, or dependent on each other.

Marginal Probability = Simple Probability , a probability of an independent event occurring.

The term “marginal variable” is used to refer to a subset of variables in a table. They called “marginal” because they used to be calculated by summing values in a table along rows or columns , and writing the sum in the margins of the table.

Examples of assigning probability.

Example 1. Tossing a coin.

Possible outcomes : (tails or heads).
Events (tails of heads) are mutually exclusive.
Their probabilities should be equal.

$P_t + P_h = 1$
$P_t = 0.5$
$P_h = 0.5$

Example 2. Rolling a die.

Possible outcomes: (1,2,3,4,5,6).
They are mutually exclusive, probability equal, sum() = 1
So each probability = 1/6 = 0.167 3.

Example 3. Job Search.

I get 2 emails with job descriptions every day, 10 per week.

Two of them may be a fit: 2 of 10 (0.2).

So I send an email that I am interested.
I am invited to every third position I requested (0.2/3 = 0.067)

And I get 1 position out of 10 initial interviews (0.0067).

So the probability that an incoming email will result in a job is 0.0067, or ~1%.
So, I need to consider 100 positions (this will take 10 weeks), send 20 emails asking to invite, go to 6..7 interviews, and eventually get a job.

So job search takes 2..3 months.

Rules (Or, And, etc.)

Lets A be an event.
P(A) – probabiltiy of event A
then: P(A) + P(not A) = 1

If A and B are events, then:
P(A or B) = P(A) + P(B) – P(A and B)

for mutually exclusive events P(A and B) = 0,
so P(A or B) = P(A) + P(B)

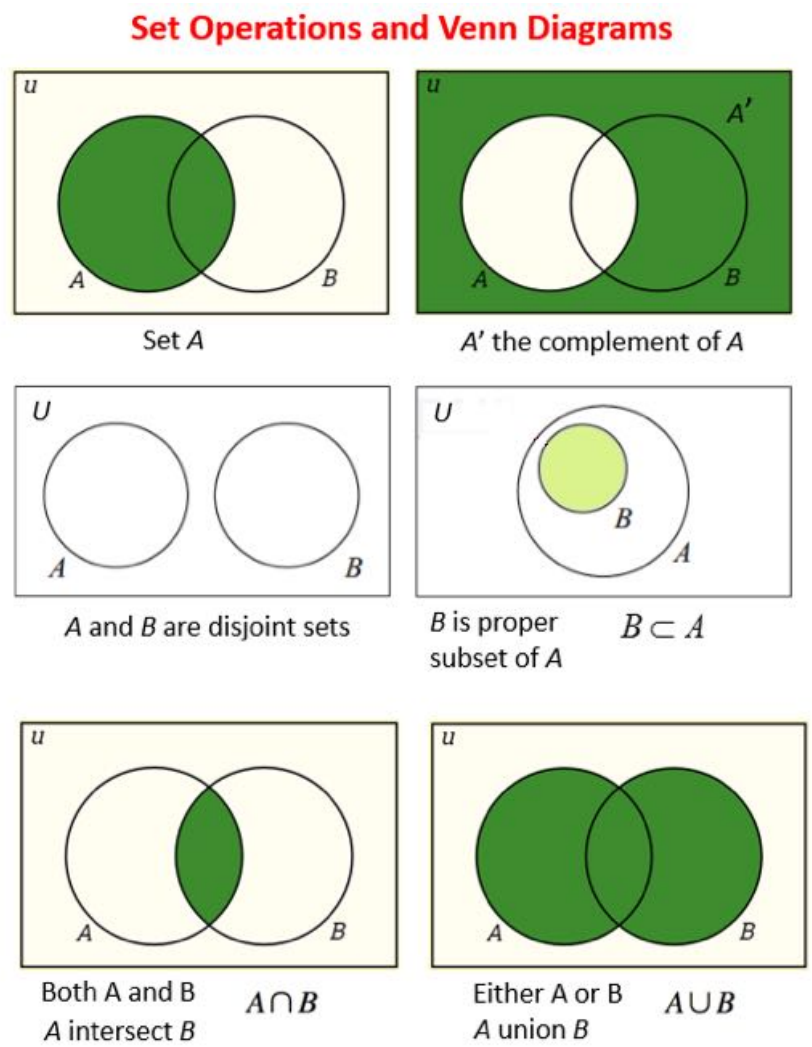
Venn Diagrams

Venn Diagrams = Graphical representation of events as circles.
See many examples – just google images for: Venn Diagram A or B.

For example, this image shows Intersection (A and B),
negation of intersection (A but not B, or B but not A),
union (A or B), and subtraction (A but not B) :

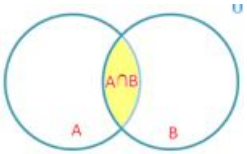
Conditional Probability
P(A|B) – probability of A given that B has occurred.
P(A|B) = P(A and B) / P(B) (if P(B) is not zero)

If A & B are independent form each other, then:
P(A|B) = P(A)
P(B|A) = P(B)



Union Bound Formula

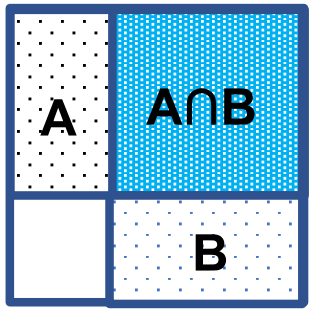
$P(A \cup B) = P(A) + P(B) - P(A \cap B) \leq P(A) + P(B)$



Bonferroni Inequality

$P(A \cap B) = P(A) + P(B) - P(A \cup B) >= P(A) + P(B) - 1$

A = smart
B = beautiful
If P(A) ~ 0.9, and P(B) ~ 0.9, then P(A & B) >= 0.8



Carlo Emilio Bonferroni
1892 – 1960
Florence, Italy

Bonferroni Correction

When conducting multiple tests, the chance of getting a significant (type 1) error increases by pure chance.
To correct for this, we proportionally "decrease" the "alpha" value of each test.

Example:
Suppose we have Ng = 5 treatment groups.
Hull hypothesis is that group members are "equal" in all groups.
Alpha = 0.05 ("family alpha rate") – chance of rejecting H0

We decided to make 10 comparisons:
1 to 2, 2 to 3, 3 to 4, 4 to 5, 1 to 3, 2 to 4, 3 to 5, 4 to 1,
(1 & 2) to (3 & 4), (1 & 2) to (4 & 5)

We will use new corrected alpha Alpha_new = Alpha/10 = 0.005 for each of those 10 tests.
If any of the 10 tests rejects the H0 (using this small Alpha_new value of 0.005),
then we reject H0 for the whole "family".

Bayesian inference - calculate probability following chains of dependencies or correlations

Bayes' Theorem (conditional probability)

$$P(A|B) = \frac{P(B|A) * P(A)}{P(B)}$$

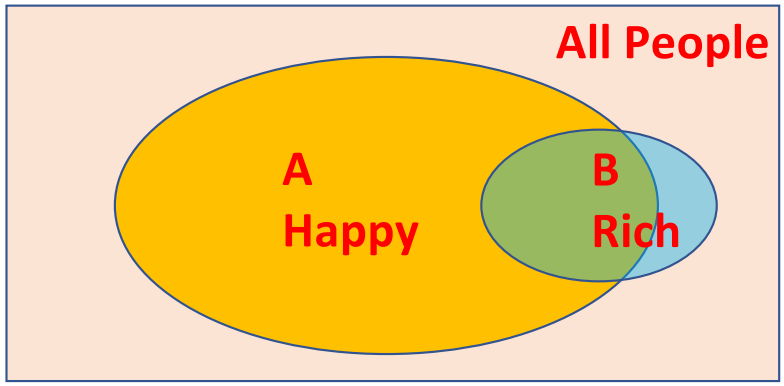
or

$$P(A|B) * P(B) = P(B|A) * P(A)$$

where A and B are events, and P(B) not equal 0.

$$\text{posterior} = \frac{\text{prior} \times \text{likelihood}}{\text{evidence}}$$

Thomas Bayes
(1701–1761)



Intuitive explanation of Bayesian rule (on the left).
A = Happy, B = Rich
P(A) = 40% = probability of being happy (Yellow)
P(B) = 5% = probability of being rich (Blue)
P(A & B) = 4% = probability of rich & happy (Green)
P(B|A) = 10% = probability of being rich given that Happy
P(A|B) = 80% = probability of being happy given that rich

You can calculate green intersection using two methods:
P(A|B)*P(B) = 80%*5% = 4%
P(B|A)*P(A) = 10%*40% = 4%

Bayes' Theorem (Law, Rule)
– https://en.wikipedia.org/wiki/Thomas_Bayes -
– https://en.wikipedia.org/wiki/Bayes%27_theorem –
– https://en.wikipedia.org/wiki/Bayesian_inference –

P(A|B) = P(B|A)*P(A)/P(B)

where A and B are events and P(B) not equal 0.

Frequentist interpretation of Bayes' theorem.
Probability measures a proportion of outcomes.
Suppose an experiment is performed many times.

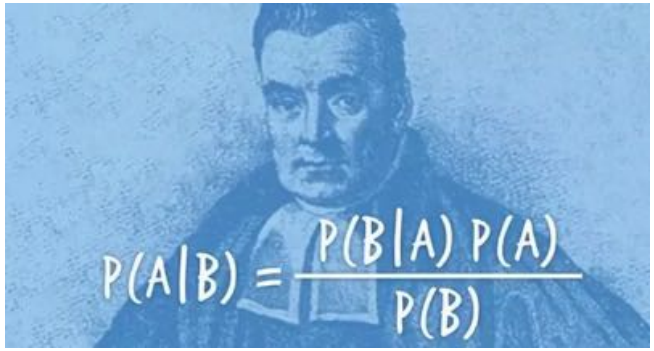
P(A) is the proportion of outcomes with property A,
and P(B) that with property B.

P(B|A) is the proportion of outcomes with property B
out of outcomes with property A,

and P(A|B) the proportion of those with A out of those with B.

Bayesian interpretation of Bayes' theorem.
Probability measures a degree of belief.
Bayes' theorem then links the degree of belief in a proposition before (A)
and after (B) accounting for evidence.

For example, suppose it is believed with 50% certainty
that a coin is twice as likely to land heads than tails.
If the coin is flipped a number of times and the outcomes observed,
then that degree of belief may rise, fall or
remain the same depending on the results.



Thomas_Bayes – (/'be?z/; 1701–1761)

For proposition A and evidence B,
P(A), the prior, is the initial degree of belief in A.
P(A|B), the posterior, is the degree of belief having accounted for B.
P(B|A)/P(B) – represents the support B provides for A.

See also Bayesian inference.
– https://en.wikipedia.org/wiki/Bayesian_inference

Bayesian inference is a method of statistical inference in which
Bayes' theorem is used to update the probability for a hypothesis
as more evidence or information becomes available.

Probability - Bayesian vs Frequentist

The **Frequentist** see probability as something that has to do with a limiting frequency based on an observed proportion.

The **Bayesian** is subjective and uses a priori beliefs to define a prior probability distribution on the possible values of the unknown parameters.

Frequentists accuse Bayesians for being subjective in selecting a prior model / distribution.

Bayesians snap back that at least they have a model, whereas Frequentists just calculate averages of observed data

Allen Downey – https://www.youtube.com/watch?v=R6d-AbkhBQ8

Frequentist (classical):

Calculate means (estimates) and confidence intervals
Problems treating missing data (**survivorship bias**)
Problems with hypothesis testing (**p-hacking**)

Bayesian:

Enumerate all possible hypotheses – and compute their probabilities.
We make a (subjective) choice of "prior" (model, distribution)
and produce posterior distribution.
Allows for complex decision analysis under uncertainty.

Example1: Polling, two candidates: Alice & Bob.
52% of polled voters intend to vote for Alice.
Confidence interval [42%,62%], p-value = 0.34
What is the probability that Alice will win ?
Classical frequentist approach can not answer this question.
Bayesian method – result is a posterior distribution of possible outcomes based on data.
Thus you can get your answer.

Example2: comparing two drugs A & B.
A is better than B with p value < 0.001
A is more expensive.
Which drug should we prescribe?
Cost-benefit.
Suppose we get predicted probability of survival with A and B.
We could then compute dollars per life saved.

Example3: Guessing game
You see the prize – and guess the price
Opponent sees a different prize – and guesses the price.
Whoever is closer (without going over) wins.

Example4: MCMC (Markov-Chain Monte-Carlo)

Bayesian vs Frequentist – Survivorship Bias in World War 2

During World War II the US military's concluded that the most-hit areas of the plane needed additional armor.
But statistician Abraham Wald (Statistical Research Group (SRG) at Columbia University) has recommended adding armor to the areas that showed the least damage. His reasoning was that the sample didn't include the planes which were shot down.

It is dangerous to use data "as is" to create a model,
because the data may be biased to start with,
and it can easily lead to creating a biased model

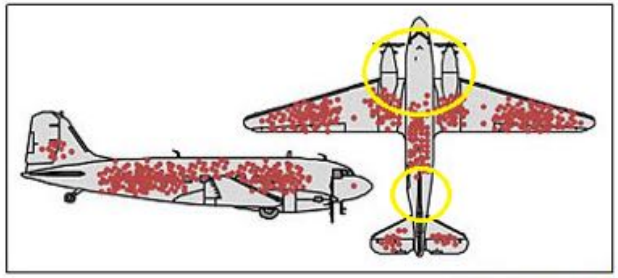
Problem occur when we have:
- under-represented features (missing data)
- over-represented features,
- "target leakage" effect.

Survivorship bias:
- https://www.youtube.com/watch?v=B3YQJ5DwTzM

Gender Bias:
- https://www.youtube.com/watch?v=p60-Jdq754A

Bias in criminal justice:
- https://www.youtube.com/watch?v=Gi4YeRqfb24

Target Leakage:
- https://www.youtube.com/watch?v=y8qaI5mpJeA



Credit: Cameron Moll

Gentlemen, you need to put more armour-plate where the holes aren't because that's where the holes were on the airplanes that didn't return - Abraham Wald 1942.

Boy or Girl Paradox

(a.k.a. "The Two Child Problem", "Mr. Smith's Children", "Mrs. Smith Problem").

Initially formulated at 1959, when Martin Gardner featured it in his October 1959 "Mathematical Games column" in Scientific American. He titled it The Two Children Problem, and phrased the paradox as follows:

Question 1: Mr. Jones has two children. The older child is a girl. What is the probability that both children are girls?

P = 1/2

Older child	Younger child
Girl	Girl
Girl	Boy
~~Boy~~	~~Girl~~
~~Boy~~	~~Boy~~

Question 2: Mr. Smith has two children. At least one of them is a boy. What is the probability that both children are boys?

P = 1/3

Older child	Younger child
~~Girl~~	~~Girl~~
Girl	Boy
Boy	Girl
Boy	Boy

Older child	Younger child	P(this family)	P(ALOB given this family)	P(ALOG given this family)	P(ALOB and this family)	P(ALOG and this family)
Girl	Girl	$\frac{1}{4}$	0	1	0	$\frac{1}{4}$
Girl	Boy	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{8}$	$\frac{1}{8}$
Boy	Girl	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{8}$	$\frac{1}{8}$
Boy	Boy	$\frac{1}{4}$	1	0	$\frac{1}{4}$	0

$$\frac{\mathrm{P(ALOB\ and\ BB)}}{\mathrm{P(ALOB)}} = \frac{\frac{1}{4}}{0 + \frac{1}{8} + \frac{1}{8} + \frac{1}{4}} = \frac{1}{2}.$$

- https://en.wikipedia.org/wiki/Boy_or_Girl_paradox

video - This May Be The Most Counterintuitive Probability Paradox I've Ever Seen | Can you spot the error?
- https://www.youtube.com/watch?v=bDZieLmya_I

video - The Boy or Girl Probability Paradox Resolved
- https://www.youtube.com/watch?v=EIB350w8iJo

Random Variable. Probability Distribution

Random variable is a numerical variable taking different values. Values may be **discrete or continuous**.

For example, think of variable “x” which may have any real value between -10 and +10.

Or think of a variable “z” which can have discrete values 0,0.1,0.2, … 0.9, 1.0

Discrete Case

Probability function P(x) of a discrete random variable “x” is simply a function which gives probability of discrete values.

For example, if the random variable “x” may have two values: (0 or 1), and if probabilities of those values are equal, then

P(0) = 0.5, f(1) = 0.5.

The sum of individual probabilities for all possible values should be 1.

Expected value is an weighted average of all values.

E(x) = sum(Xi*P(Xi)), where Xi goes through all allowed values of X.

Variance = sigma squared = sum((Xi-E(x))**2 * P(Xi))

Standard Deviation = sqrt(sigma squared)

Probability Density Function (PDF)
vs. Cumulative Probability Function (CPF)

Continuous Case

Probability Density Function (PDF) f(x) is a continuous function.

f(x)*dx gives probability that value is between x and x+dx.

The total integral of f(x)*dx should be equal to 1.

Cumulative Probability Function (CPF)
This is simply a function F(x) which gives probability that value is less than x. It monotonously grows from 0 to 1 as x grows (from small to large values).

The PDF is a first derivative of CPF.

CPF(x+dx) – CPF(x) = = [Prob. of value < (x+dx)] – [Prob. of value < x] = f(x)*dx

Mean value = Expected value

For discrete case we had:
E(x) = sum(Xi*P(Xi))

For continuous case this translates into
E(x) = Integral(xF(x)dx)